MetalLB + Traefik

Kubernetes Homelab Configuration & Operations Guide

Environment: Rocky Linux Kubernetes • Argo CD GitOps • NGINX test application • 11 October 2026

1. Overview and verified environment

Goal: route HTTP requests from a home-network client through a stable LAN address to Kubernetes applications without exposing an application-specific NodePort. MetalLB allocates and advertises a LAN IP for the Traefik LoadBalancer Service. Traefik processes Ingress rules and forwards requests to the nginx-web ClusterIP Service.

Component	Observed configuration
Kubernetes node LAN	192.168.178.49 (master)
Traefik namespace	traefik
Traefik LoadBalancer IP	192.168.178.240
Traefik Service ClusterIP	10.109.145.174
Traefik HTTP / HTTPS	80:31353/TCP and 443:30273/TCP
IngressClass	traefik → traefik.io/ingress-controller
Application hostname	nginx.home.lab
Backend Service	nginx-web:80 (ClusterIP)
Git repository	https://github.com/Syeem007/manifest.git (app1)

2. Traffic architecture

```
LAN browser / curl
  | DNS or hosts: nginx.home.lab -> 192.168.178.240
  v
MetalLB-advertised VIP: 192.168.178.240:80
  | (Service type LoadBalancer)
  v
Traefik controller (namespace: traefik)
  | IngressClass: traefik; Host: nginx.home.lab
  v
Ingress: nginx-web-ingress (same namespace as backend)
  v
Service: nginx-web (ClusterIP, port 80)
  v
NGINX Pods (dynamic Pod IPs)
```

3. Install MetalLB with Helm

Run these commands on the Kubernetes control-plane host with Helm and kubectl configured. Use the Helm chart defaults appropriate to your cluster; validate chart options against the installed version.

```
kubectl create namespace metallb-system
kubectl label namespace metallb-system \
  pod-security.kubernetes.io/enforce=privileged \
  pod-security.kubernetes.io/audit=privileged \
  pod-security.kubernetes.io/warn=privileged
helm repo add metallb https://metallb.github.io/metallb
helm repo update
helm upgrade --install metallb metallb/metallb \
  --namespace metallb-system
kubectl get pods -n metallb-system
```

MetalLB requires a suitable CNI/network environment and working speaker Pods. In some clusters, strict ARP or kube-proxy configuration may require additional checks.

4. Configure the MetalLB address pool

Important: the example 192.168.178.240–245 range must be unused, within the local subnet, and excluded/reserved outside the router DHCP allocation range. The observed assigned address was 192.168.178.240.

```
# metallb-config.yaml
apiVersion: metallb.io/v1beta1
kind: IPAddressPool
metadata:
  name: home-lan-pool
  namespace: metallb-system
spec:
  addresses:
    - 192.168.178.240-192.168.178.245
---
apiVersion: metallb.io/v1beta1
kind: L2Advertisement
metadata:
  name: home-lan-advertisement
  namespace: metallb-system
spec:
  ipAddressPools:
    - home-lan-pool
```

```
kubectl apply -f metallb-config.yaml
kubectl get ipaddresspools,l2advertisements -n metallb-system
kubectl get svc traefik -n traefik
```

5. Install Traefik with Helm

Traefik runs separately from the application. Its LoadBalancer Service is the common HTTP/HTTPS entry point. If Traefik is already installed, do not install a second release.

```
helm repo add traefik https://traefik.github.io/charts
helm repo update
helm upgrade --install traefik traefik/traefik \
  --namespace traefik --create-namespace
```

```
kubectl get pods -n traefik
kubectl get svc traefik -n traefik
kubectl get ingressclass
```

Observed result: Traefik Pod Running; Service type LoadBalancer; external IP 192.168.178.240; IngressClass traefik. A bare-metal LoadBalancer may show <pending> before MetalLB is installed/configured.

6. Application Service and Ingress

Place the Ingress and nginx-web Service in the same namespace. Example below uses default; if the application is deployed elsewhere, change both namespace values consistently. Do not use ingress-nginx merely because it is the name of an old controller.

```
# service.yaml
apiVersion: v1
kind: Service
metadata:
  name: nginx-web
  namespace: default
spec:
  type: ClusterIP
  selector:
    app: nginx-web
  ports:
    - port: 80
      targetPort: 80
      protocol: TCP
---
# ingress.yaml (save as separate file when applying)
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: nginx-web-ingress
  namespace: default
spec:
  ingressClassName: traefik
  rules:
    - host: nginx.home.lab
      http:
        paths:
          - path: /
            pathType: Prefix
            backend:
              service:
                name: nginx-web
                port:
                  number: 80
```

The Service selector app=nginx-web must match the Pod labels in the Deployment. The IngressClass must be traefik (not nginx). The Traefik Helm installation already creates the class; do not manage a duplicate class manually in the application repository.

7. Argo CD GitOps deployment

In Argo CD, use repository https://github.com/Syeem007/manifest.git, revision main, path app1, and destination namespace matching the manifests (for example default). Enable automated sync if desired. Commit the application Deployment, Service and Ingress manifests to Git. Keep Helm-owned Traefik IngressClass separate from the application.

```
kubectl get applications -n argocd
kubectl get pods,svc,ingress -n default
kubectl describe ingress nginx-web-ingress -n default
```

8. Validate end-to-end routing

First test with an explicit Host header; this bypasses DNS and proves the routing path. Replace IPs if your LoadBalancer allocation changes.

```
curl -v -H "Host: nginx.home.lab" http://192.168.178.240/
# Expected: HTTP/1.1 200 OK and the NGINX welcome page
kubectl get svc traefik -n traefik
kubectl get ingress -A -o wide
```

For browser access on macOS, map nginx.home.lab to 192.168.178.240 using local DNS or /etc/hosts, then open http://nginx.home.lab. The earlier successful test through 192.168.178.49:31353 with Host header proved Traefik routing before MetalLB was added. The new LoadBalancer IP should be tested separately from the Mac.

```
# On Mac, add to /etc/hosts (requires sudo):
192.168.178.240 nginx.home.lab
# Then test:
curl -I http://nginx.home.lab/
```

9. Troubleshooting

Symptom	What to check
External IP <pending>	MetalLB Pods, address pool, L2Advertisement, free LAN IP, speaker logs
Traefik returns 404	Ingress exists; correct host, namespace and ingressClassName: traefik
Traefik returns 502/503	Backend Service selector, endpoints, Pod readiness, network policies
Connection timeout from Mac	Mac LAN reachability, node/firewall rules, MetalLB L2 advertisements
Hostname does not resolve	Mac /etc/hosts or local DNS record
Ingress missing after namespace deletion	Argo CD application destination and sync status; recreate desired manifests in an existing namespace

```
kubectl get pods -n metallb-system
kubectl logs -n metallb-system -l component=speaker --tail=100
kubectl get svc traefik -n traefik -o wide
```

```
kubectl logs -n traefik deployment/traefik --tail=100
kubectl get ingress -A
kubectl get endpoints nginx-web -n default
```

10. LAN vs public internet

192.168.178.240 is a private RFC1918 LAN address. MetalLB does NOT make it reachable from the public internet. Public access requires a publicly reachable entry point, such as a secure tunnel, or a router with usable public IP plus carefully configured forwarding, public DNS, TLS and firewall controls. CGNAT/DS-Lite can block direct IPv4 inbound access. Avoid exposing Jenkins, Argo CD or the Kubernetes API directly.

11. Operational notes

MetalLB handles IP allocation/advertisement; Traefik handles HTTP routing; Ingress declares host/path rules; Service selects Pods. Pod IPs can change without changing the client URL. Service ClusterIPs may change if Services are deleted/recreated. Store declarative manifests in Git and avoid manual edits to Argo CD-managed resources.